EXAM REVISION 1 – SOLUTIONS
GRADE 10 PAPER 2
MATHEMATICS

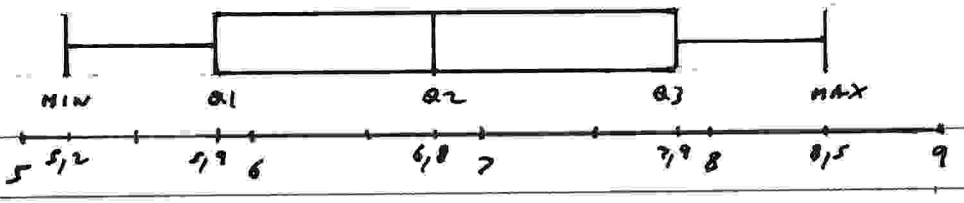
QUESTION 1

1.1.1 17 1.1.2 5,7 AND 6,3 1.1.3 $\bar{x} = 6,8$

1.1.4 $Q2 = 6,8$ 1.1.5 $Q1 = \frac{5,7+6,0}{2} = 5,85$ $Q3 = \frac{7,7+8,0}{2} = 7,85$

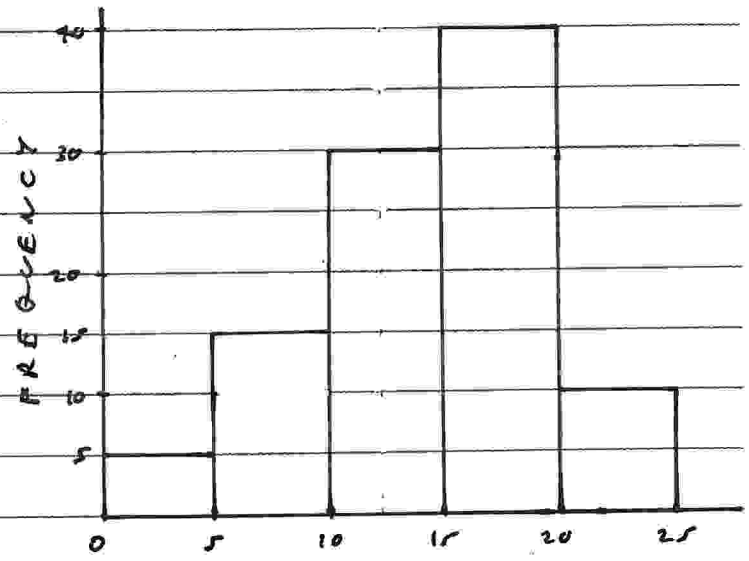
$\therefore IQR = Q3 - Q1 = 2$

1.1.6



1.1.7 50%

1.2.1



1.2.2 POSITION = $101/4 \approx 25$

$\therefore (10 ; 15]$

1.2.3 $(2,5 \times 5 + 7,5 \times 15 + 12,5 \times 30 + 17,5 \times 40 + 22,5 \times 10) \div 100$

$\therefore \bar{x} = 14,25$ %

1.2.4 MIDPOINTS OF THE CLASSES ARE USED % OF SALARY

SINCE THE EXACT INDIVIDUAL DATA IS NOT AVAILABLE.

1.3.1 $90 - 35 = 55$ 1.3.2 $SIQR = (65 - 45)/2 = 10$

1.3.3 25% 1.3.4 50% → 125 ∴ 250 WROTE

QUESTION 2

2.1.1 $x = 1000 - 700 = 300$ 2.1.2 $60 < S \leq 80$

2.1.3 POSITION $= \frac{3}{4}(1001) \approx 750$ ∴ $80 < S \leq 100$

2.1.4 $219 + 30 + 3 = 252$

2.1.5 $(50 + 1350 + 15000 + 27860 + 19710 + 3300 + 390) \div 1000$

$\bar{x} = 68$ Km/h

2.2.1 $750 - 850$

2.2.2 $1000 + 3600 + 8400 + 12800 + 12600 + 8000 + 4400 = 50800$

2.2.3 62 WEEKS

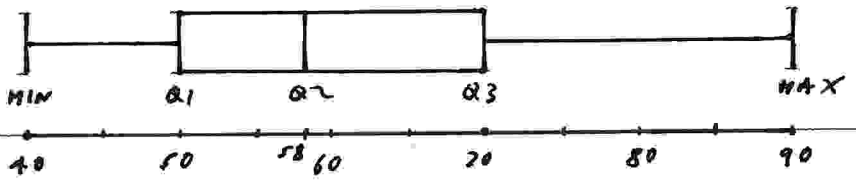
2.2.4 $50800 \div 62 = 820$

2.2.5 $62 \div 2 = 31$: $2+6=8 \rightarrow 8+12=20 \rightarrow 20+16=36$ ✓ $\therefore$ 800

2.2.6 $63 \div 4 = 15{,}75$: $2+6=8 \rightarrow 8+12=20$ ✓ $\therefore$ 650 – 750

2.3 40 44 48 50 52 54 56 60 62 66 70 70 76 90

↑ MIN ↑ Q1 ↑ Q2 = 58 ↑ Q3 ↑ MAX



2.4 $72 - a = 31 \rightarrow a = 41$ $\quad d = 72$

$(39 + 41 + b + 52 + b + 16 + 72 + 72)/7 = 56$

$\therefore 2b = 100 \rightarrow b = 50$ $\quad \therefore c = 66$

2.5 $c = d = 15$, $e = 19$, $b = 10$, $a = 9$

$\therefore$ MAXIMUM $(a+b+c+d+e) = 112$

QUESTION 3

3.1 $m_{AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{k+1}{-4}$ AND $m_{BC} = \frac{-1+k}{-3}$ $\therefore \frac{k+1}{-4} = \frac{k-1}{-3}$

$-3k - 3 = -4k + 4 \rightarrow k = 7$

3.2 $m_{AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-11+2}{-3} = 3$ AND $m_{BC} = \frac{12}{t}$ $\therefore \frac{12}{t} = 3 \rightarrow t = 4$

$m_{DB} = \frac{n+2}{9}$ $\therefore \frac{n+2}{9} = -\frac{1}{3} \rightarrow 3n + 6 = -9 \rightarrow n = -5$

3.3.1 $1 = \frac{n-4}{2} \rightarrow n = 6$ AND $m = \frac{5-3}{2} \rightarrow m = 1$

3.3.2 $MP^2 = (t-1)^2 + (-2-1)^2 \rightarrow 58 = t^2 - 2t + 1 + 9$

$t^2 - 2t - 48 = 0 \rightarrow (t-8)(t+6) = 0 \rightarrow t = 8$ OR $-6 \rightarrow t = 8$

3.3.3 $AM^2 = (1+4)^2 + (1+3)^2 = 41 \rightarrow AM = \sqrt{41}$ $\therefore$ RADIUS $= \sqrt{41}$ UNITS

3.4.1 $0 = \frac{n-4}{2} \rightarrow n = 4$ AND $3 = \frac{m+5}{2} \rightarrow m = 1$

3.4.2 $A(-4;1)$; $B(4;5) \rightarrow m_{AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5-1}{4+4} \rightarrow m_{AB} = \frac{1}{2}$

3.4.3 $m_{AC} = -2 \rightarrow AC: y - 1 = -2(x+4) \rightarrow y = -2x - 7$

$\therefore C(0;-7)$ $\therefore MC = 10$ UNITS

3.4.4 AREA $\Delta AMC = \frac{1}{2} . 10 . 4 = 20$ UNITS^2

3.4.5 $BD^2 = (3-4)^2 + (k-5)^2 \rightarrow 50 = 1 + k^2 - 10k + 25$

$k^2 - 10k - 24 = 0 \rightarrow (k-12)(k+2) = 0 \rightarrow k = -2$

3.5.1 TRAPEZIUM

3.5.2 D: $y = 3x \rightarrow 9 = 3x \rightarrow x = 3 \rightarrow D(3;9) \rightarrow DB = 9 - 3 = 6$

$\therefore$ AREA $ODBA = \frac{1}{2} . 9(9+6) = 67,5$ UNITS^2

3.5.3 $m_{OD} = 3 \rightarrow m_{AP} = -\frac{1}{3}$ $AP: y - 0 = -\frac{1}{3}(x-9)$

$y = -\frac{1}{3}x + 3$

P: $3x = -\frac{1}{3}x + 3 \rightarrow 9x = -x + 9 \rightarrow x = \frac{9}{10} \rightarrow y = \frac{27}{10}$

$\therefore P\left(\frac{9}{10} ; \frac{27}{10}\right)$

QUESTION 4

4.1.1 $m_{AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6-0}{0+3} = 2 \rightarrow m_{BC} = -\frac{1}{2}$

$\therefore -\frac{1}{2} = \frac{-\frac{5}{2} - 0}{t+3} \rightarrow -t - 3 = -5 \rightarrow -t = -2$

$\therefore t = 2$

4.1.2.1 $D\left(3;-\frac{5}{2}\right)$ (ANY $D\left(x>2;-\frac{5}{2}\right)$)

4.1.2.2 $D\left(9;-\frac{5}{2}\right)$

4.2.1 $m_{CD}=\frac{y_2-y_1}{x_2-x_1}=\frac{-1+3}{6-2}=\frac{2}{4}\rightarrow m_{CD}=\frac{1}{2}$

4.2.2 $m_{BC}=\frac{5+1}{2-6}=-\frac{3}{2}\rightarrow BC: y+1=-\frac{3}{2}(x-6)\rightarrow y=-\frac{3}{2}x+8$

4.2.3 $m_{AP}=\frac{1}{2}\rightarrow AP: y-2=\frac{1}{2}(x+1)\rightarrow y=\frac{1}{2}x+\frac{5}{2}$

$P: \frac{1}{2}x+\frac{5}{2}=-\frac{3}{2}x+8\rightarrow x+5=-3x+16\rightarrow 4x=11\rightarrow x=\frac{11}{4}$

$\therefore y=\frac{31}{8}\rightarrow P\left(\frac{11}{4};\frac{31}{8}\right)$

4.2.4 $BD\ //\ y\text{-AXIS}\rightarrow BD=8$ UNITS

$\therefore$ AREA $ABCD=\frac{1}{2}.8.3+\frac{1}{2}.8.4=28$ UNITS^2

4.3.1.1 $PA^2=(-6-1)^2+(t-2)^2\rightarrow 58=49+t^2-4t+4\rightarrow t^2-4t-5=0$

$(t-5)(t+1)=0\rightarrow t=5$ OR $t=-1\quad\therefore t=5$

4.3.1.2 $\frac{3}{2}=\frac{-4-2}{n-1}\rightarrow 3n-3=-12\rightarrow 3n=-9\rightarrow n=-3$

4.3.1.3 $m_{PD}=\frac{y_2-y_1}{x_2-x_1}=\frac{2+6}{1-3}=-4\rightarrow -4=\frac{-2k-2}{k-1}$

$\therefore -4k+4=-2k-2\rightarrow -2k=-6\rightarrow k=3$

4.3.1.4 LET MIDPOINT BE M: $\rightarrow x_M=\frac{-3+1}{2}=-1$ AND $y_M=\frac{2-4}{2}=-1$

$\therefore M(-1;-1)$

$m_{PQ}=-\frac{2}{3}\rightarrow PQ: y+1=-\frac{2}{3}(x+1)\rightarrow y=-\frac{2}{3}x-\frac{5}{3}$

x-INT: $y=0\rightarrow \frac{2}{3}x=-\frac{5}{3}\rightarrow x=-\frac{5}{2}\rightarrow Q\left(-\frac{5}{2};0\right)$

$R\left(0;-\frac{5}{3}\right)$

$\therefore$ AREA $\Delta QOR=\frac{1}{2}.\frac{5}{2}.\frac{5}{3}=\frac{25}{12}=2{,}08$ UNITS^2

4.4.1 $m_{AC} = \frac{8-0}{-2-6} = -1$ AND $m_{BD} = \frac{12+4}{10+6} = 1 \rightarrow m_{AC} \times m_{BD} = -1$
$\therefore A\hat{Q}D = 90°$

4.4.2 $AD^2 = (10+2)^2 + (12-8)^2 = 160 \rightarrow AD = \sqrt{160}$
$BC^2 = (6+6)^2 + (0+4)^2 = 160 \rightarrow BC = \sqrt{160}$ $\therefore AD = BC$

4.4.3 ABCD IS A RHOMBUS. ABCD IS A //GRAM SINCE ONE PAIR OF OPPOSITE SIDES ARE EQUAL & PARALLEL. AND DIAGONALS ARE ⊥.

4.4.4 LET M BE THE MIDPOINT OF CD. $\rightarrow x_M = \frac{10+6}{2} = 8$; $y_M = \frac{12+0}{2} = 6$
$\therefore M(8;6)$ $\therefore$ C IS NOW MIDPOINT OF MP.
$\therefore 6 = \frac{a+8}{2} \rightarrow a = 4$ AND $0 = \frac{b+6}{2} \rightarrow b = -6$

4.4.5.1 $E(-3;8)$ (ANY $E(x<-2;8)$)
4.4.5.2 $E(-6;8)$ 4.4.5.3 $E(-14;8)$

QUESTION 5

5.1.1 $\cos(37+23) = \cos 60° = 0,5$ $\therefore$ TRUE
5.1.2 $\sin(37-23) = \sin 14° = 0,24$ AND $\sin 37 - \sin 23 = 0,21$ $\therefore$ FALSE
5.1.3 $\sin^2 37 + \cos^2 23 = 1,21$ $\therefore$ FALSE
5.1.4 $\cos 46 = 0,69$ AND $2.\cos^2 23 - 1 = 0,69$ $\therefore$ TRUE
5.1.5 $\sin 74 = 0,96$ AND $2.\sin 37.\cos 37 = 0,96$ $\therefore$ TRUE
5.1.6 $\tan 60° = 1,73$ AND $\sin 60 / \cos 60 = 1,73$ $\therefore$ TRUE

5.2.1 $2 \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$ 5.2.2 $(\sqrt{3})^2 - 3 = 3 - 3 = 0$

5.2.3 $\frac{1}{\cos 45} + \frac{1}{\sqrt{2}} \cdot 1 = \frac{2}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$

5.2.4 $\frac{7}{4} - \frac{1}{\sin 30} + \left(\frac{1}{2}\right)^2 = \frac{7}{4} - 2 + \frac{1}{4} = 0$

5.2.5 $\sin 30 . \frac{1}{\sin 30} . \cos 30 . \frac{\sin 30}{\cos 30} = \sin 30 = \frac{1}{2}$

OR JUST WRITE AS 1 SINCE $\sin^2 A + \cos^2 A = 1$

5.2.6 $\frac{1}{(\tan 30)^2} - 2\left(\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right) = \left(\sqrt{3}\right)^2 - 2\left(\frac{3}{4} + \frac{1}{4}\right) = 3 - 2 = 1$

5.2.7 $\sqrt{3}.1 + \sqrt{2}.\frac{\sqrt{2}}{2} - 2.\frac{\sqrt{3}}{2} = \sqrt{3} + \frac{2}{2} - \sqrt{3} = 1$

5.2.8 $2.\frac{\sqrt{2}}{2}.\frac{\sqrt{2}}{2} + \left(\frac{1}{2}\right)^2 - 1 = 1 + \frac{1}{4} - 1 = \frac{1}{4}$

5.2.9 $\frac{\sin 60}{\tan 60} = \frac{\frac{\sqrt{3}}{2}}{\sqrt{3}} = \frac{1}{2}$ $\left(\text{OR } \sin 60 . \frac{\cos 60}{\sin 60} = \cos 60 = \frac{1}{2}\right)$

5.3 $\frac{1}{\sqrt{3}}.\frac{\sqrt{3}}{2} + \sin B - 1 = 0 \rightarrow \sin B = \frac{1}{2} \quad \therefore B = 30^\circ$

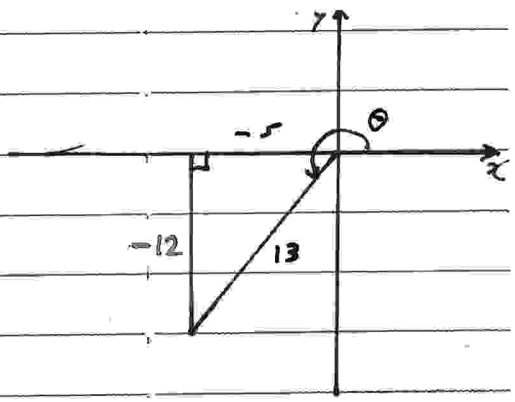
5.4.1 $T = \frac{3.\sin 55{,}8}{\cos 60 - \tan 40} = -7{,}317$

5.4.2 $T = \frac{3.\sin 150^\circ}{\cos 150^\circ - \tan 40} = -0{,}880$

5.5.1 $-0{,}1$ 5.2.2 $2{,}1$ 5.5.3 $2{,}1$

QUESTION 6

6.1.1 $\cos\theta = -\frac{5}{13} \rightarrow$ cos -ve $\Rightarrow$ Q2/Q3
TAN +ve $\Rightarrow$ Q1/Q3



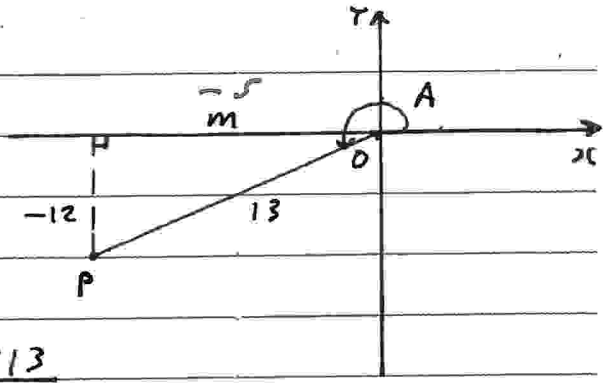
$\sin\theta = -\frac{12}{13}$

$13^2 = y^2 + 5^2$

$\therefore y = \pm 12$

6.1.2 $\sec\theta . \cot\theta = \frac{13}{-5} . \left(\frac{-5}{-12}\right) = -\frac{13}{12}$

6.1.3 $\frac{169}{144} - \operatorname{cosec}^2\theta = \frac{169}{144} - \left(\frac{13}{-12}\right)^2 = \frac{169}{144} - \frac{169}{144} = 0$

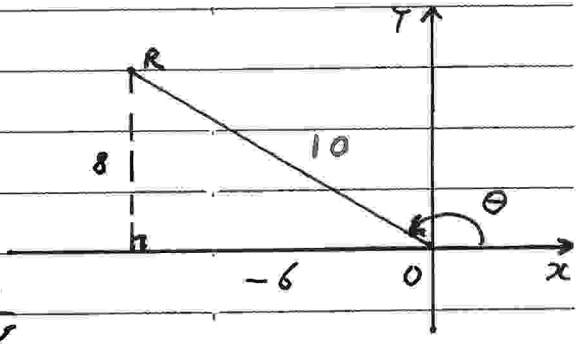


6.3.1 $13^2 = m^2 + 12^2 \rightarrow m = -5$

6.3.2 $\cos A = -\frac{5}{13}$

6.3.3 $\text{COSEC}A + \sin A = \frac{13}{-12} + \left(\frac{-12}{13}\right) = -\frac{313}{156}$

6.3.4 $\frac{1}{\cot A} - \tan A . \sec A = \frac{12}{5} - \frac{12}{5}\left(\frac{13}{-5}\right) = \frac{12}{5} + \frac{156}{25} = \frac{216}{25}$

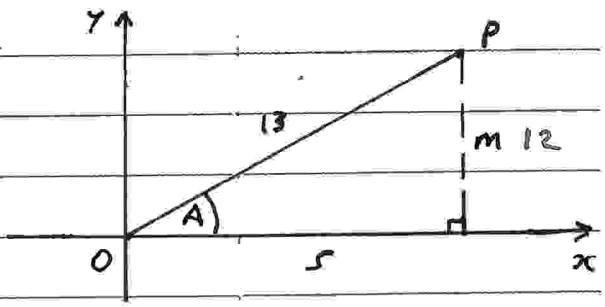


6.4.1 $OR^2 = 64 + 36 = 100 \rightarrow OR = 10$

6.4.2 $1 - 2.\frac{8}{10}.\left(\frac{-6}{10}\right) = 1 + \frac{24}{25} = \frac{49}{25}$

6.4.3 $\cos 2\theta = \left(\frac{-6}{10}\right)^2 - \left(\frac{8}{10}\right)^2 = \frac{36}{100} - \frac{64}{100} = -\frac{7}{25}$

6.4.4 $\frac{-10}{8} - \frac{6}{8} + \frac{10}{8} = \frac{-40-18+30}{24} = \frac{-28}{24} = -\frac{7}{6}$

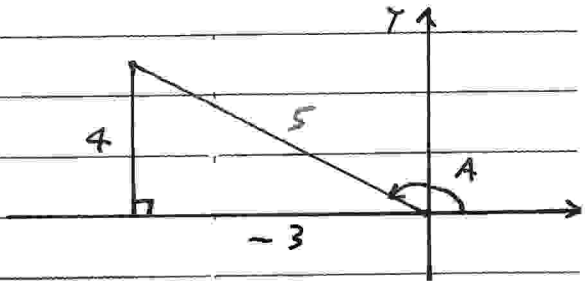


6.5.1 $13^2 = m^2 + 5^2 \rightarrow m^2 = 144 \rightarrow m = 12$

6.5.2.1 $13.\left(\frac{12}{13}\right) - 2 = 12 - 2 = 10$

6.5.2.2 $\left(\frac{5}{13}\right)^2 - 1 = \frac{25}{169} - 1 = -\frac{144}{169}$

6.5.2.3 $\frac{13}{5} + \frac{12}{5} = \frac{25}{5} = 5$



6.6.1 $\tan A = -\frac{4}{3}$: $\tan < 0 \rightarrow Q2/Q4$

$\therefore$ (Q2)

$\therefore \sin A = \frac{4}{5}$

6.6.2 $\sec A = -\frac{5}{3}$

6.6.3 $\frac{-\frac{3}{4}}{1-\left(\frac{5}{4}\right)^2} = -\frac{3}{4}\left(\frac{16}{16-25}\right) = \frac{4}{3}$

$r^2 = 16 + 9 = 25$

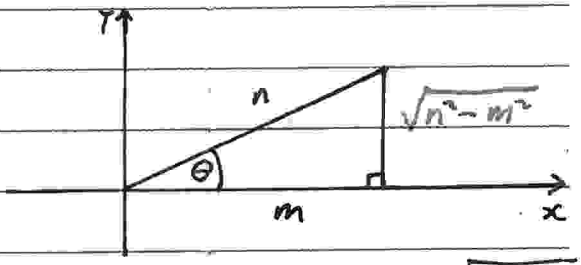
$\therefore r = 5$

6.7.1 $\sin B > 0 \rightarrow Q1/Q2$ AND $\cos B < 0 \rightarrow Q2/Q3$ $\therefore Q2$

6.7.2 $\cos B > 0 \rightarrow Q1/Q4$ AND $\tan B < 0 \rightarrow Q2/Q4$ $\therefore Q4$

6.7.3 $\sin B \cdot \tan B > 0 \rightarrow (+)(+)$ OR $(-)(-)$ $\therefore Q1$ OR $Q4$

6.8.1 $\operatorname{cosec}\theta = \dfrac{n}{\sqrt{n^2-m^2}}$



6.8.2 $m^2\left(\dfrac{n^2-m^2}{m^2}\right) + n^2\left(\dfrac{n^2-m^2}{n^2}\right) = 2n^2-2m^2$

$n^2 = r^2+m^2 \rightarrow r = \pm\sqrt{n^2-m^2}$

6.9.1 $W(\text{MAX.}) = 1-0 = 1$ 6.9.2 $W(\text{MIN.}) = 1-4 = -3$

6.9.3 $0 = 1-4\sin^2\theta \rightarrow \sin^2\theta = \dfrac{1}{4} \rightarrow \sin\theta = \dfrac{1}{2}$ $\therefore \theta = 30°$

QUESTION 7

7.1.1 $\sin 2x = 1/3 \rightarrow 2x = 19,47 \rightarrow x = 9,74°$

7.1.2 $1/2x + 40° = 75,96 \rightarrow \frac{1}{2}x = 35,96 \rightarrow x = 71,93°$

7.1.3 $\cos(x-5°) = 2/\sqrt{5} \rightarrow x-5 = 26,57 \rightarrow x = 31,57°$

7.1.4 $\sin(x/3) = \sqrt{3}/4 \rightarrow x/3 = 25,66 \rightarrow x = 76,98°$

7.1.5 $\cos x = 2,5/\pi \rightarrow x = 37,27°$

7.1.6 $\sin(x+25) = 4/3\sqrt{3} \rightarrow x+25 = 50,34 \rightarrow x = 25,34°$

7.1.7 $\cos 3x = (\tan 80)/6 \rightarrow 3x = 19,05 \rightarrow x = 6,35°$

7.1.8 $(\sqrt{7}-\sqrt{3})\sin\dfrac{x}{2} = \cos 70° \rightarrow \sin\dfrac{x}{2} = (\cos 70)/(\sqrt{7}-\sqrt{3})$

$x/2 = 21,98 \rightarrow x = 43,97°$

7.1.9 $\dfrac{1}{\sin 2x} = \sqrt{3}\cdot\cos 50° \rightarrow \sin 2x = \dfrac{1}{\sqrt{3}\cdot\cos 50°}$

$\therefore 2x = 63,92 \rightarrow x = 31,96°$

7.1.10 $\cos(x-20) = \dfrac{\sqrt{7}}{6} \rightarrow x-20 = 68,83 \rightarrow x = 83,83°$

7.2.1 $3.\sin x + \cos y = \sqrt{2}$ ① AND $2.\sin x - \cos y = 1/2$ ②

① + ②: $5.\sin x = \sqrt{2} + 1/2 \rightarrow x = 22,51° \rightarrow y = 74,59°$

OR LET $a = \sin x$, $b = \cos y \rightarrow 3a + b = \sqrt{2}$ AND $2a - b = \frac{1}{2}$, ETC

7.2.2 $0,9 - \cos y = 0,2 \rightarrow \cos y = 0,7 \rightarrow y = 45,57° ; x = 32,08°$

7.2.3 $\tan 2x - \cos 2y = 5$ ① AND $3.\cos 2y - \tan 2x = -3,5$ ②

① + ②: $2.\cos 2y = 1,5 \rightarrow y = 20,70° ; x = 40,07°$

OR LET $a = \tan 2x$, $b = \cos 2y \rightarrow a - b = 5$ AND $3b - a = -3,5$ ETC.

QUESTION 8

8.1.1 $\sin\theta = 5/10 \rightarrow \sin\theta = 1/2 \rightarrow \theta = 30°$

8.1.2 $\cos 30° = BC/10 \rightarrow BC = 10.\frac{\sqrt{3}}{2} \rightarrow BC = 5\sqrt{3}$

8.1.3 $10^2 = BC^2 + 5^2 \rightarrow BC^2 = 75 \rightarrow BC = \sqrt{75} \rightarrow BC = 5\sqrt{3}$

8.2 $P\hat{C}A = 46° \rightarrow B\hat{C}A = 41° \rightarrow \tan 41 = 50/AC \rightarrow AC = 57,518m$

$\tan 46 = AP/AC \rightarrow AP = 57,518 . \tan 46 \rightarrow AP = 59,56 m$

$\therefore h = 9,56$ METRES

8.3 $B\hat{A}C = D\hat{A}C = 30° \rightarrow \cos 30 = K/AC \rightarrow AC = 2K/\sqrt{3}$

$\tan 30 = BC/AC \rightarrow BC = \frac{2.K}{\sqrt{3}} . \frac{1}{\sqrt{3}} \rightarrow BC = \frac{2}{3}K$

8.4.1 $\sin 60° = BD/18 \rightarrow BD = 18.\frac{\sqrt{3}}{2} \rightarrow BD = 9\sqrt{3}$ METRES

8.4.2 $BE = 9\sqrt{3} - 2 \rightarrow CE^2 = (9\sqrt{3} - 2)^2 + 10^2 \rightarrow CE = 16,87 m$

8.4.3 $\tan \hat{C} = \frac{9\sqrt{3} - 2}{10} \rightarrow \hat{C} = 53,6°$ (ANGLE OF ECEVATION)

8.5.1 $\sin 63 = BW/9 \rightarrow BW = 8,02$ m

8.5.2 $\cos 63 = AW/9 \rightarrow AW = 4,086$

$B_1W = 8,02 - 1,02 = 7 \rightarrow A_1W^2 = 9^2 - 7^2 \rightarrow A_1W = 5,657$

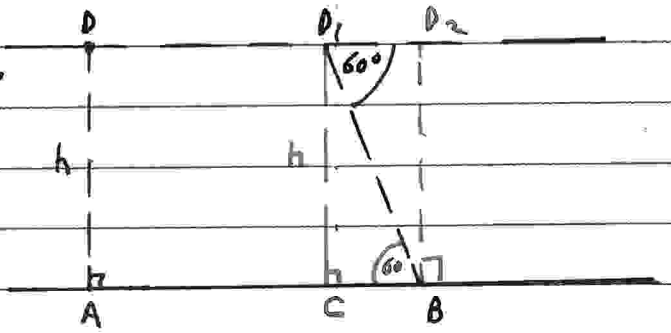
$\therefore AA_1 = 5,657 - 4,086 \rightarrow AA_1 = 1,57$ METRES

8.6.1 DRAW $D_1C \perp AB$

$\therefore BC = 54,85$ m ; $D_1\hat{B}C = 60°$

$\therefore \tan 60 = h/54,85$

$\therefore h = 95,00$ m



8.6.2 $DD_2 = 254,85$ m

$\therefore 254,85 = 15t \rightarrow t = 16,99 \quad \therefore$ 17 SECONDS

8.7.1 $P\hat{M}A = 40° \rightarrow \cos 40 = AM/20 \rightarrow AM = 20.\cos 40$

$\sin 40 = AP/20 \rightarrow AP = 20.\sin 40 \quad \therefore BQ = 40.\sin 40$

$\therefore \tan 50 = (40.\sin 40)/MB \rightarrow MB = (40.\sin 40)/\tan 50$

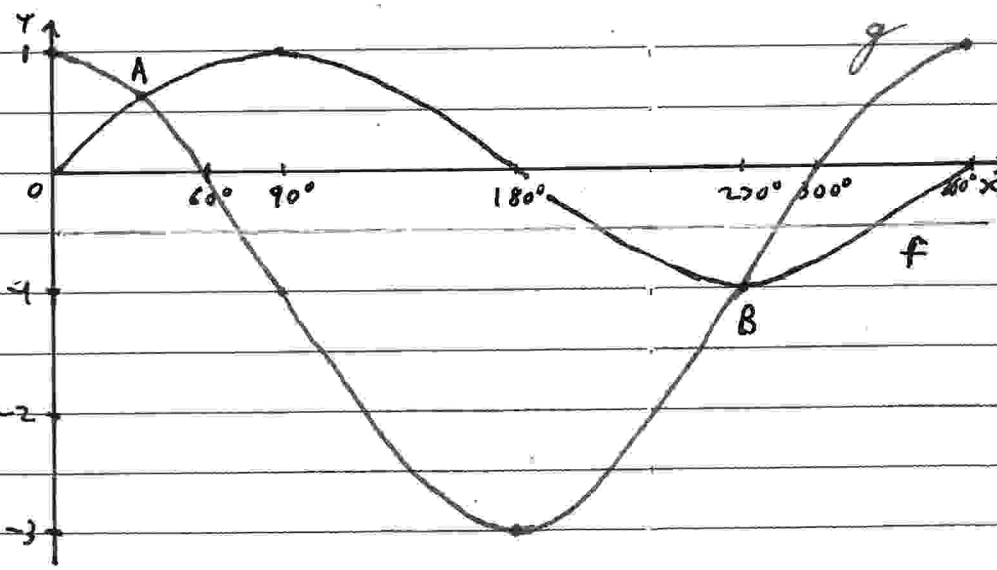
$\therefore AB = 20.\cos 40 + \dfrac{40.\sin 40}{\tan 50} \rightarrow AB = 36,90$ m

8.7.2 $P\hat{M}Q = 90° \quad \sin 50 = (40.\sin 40)/MQ \rightarrow MQ = \dfrac{40.\sin 40}{\sin 50}$

$\therefore PQ^2 = 20^2 + \left(\dfrac{40.\sin 40}{\sin 50}\right)^2 \rightarrow PQ = 39,07$ m

QUESTION 9

9.1.1 g; X-INT: $Y = 0 \rightarrow \cos x = \frac{1}{2} \rightarrow x = 60°$ OR $x = 300°$



9.1.2.1 f: 1 g: 2

9.1.2.2 $Y \in [-3; 1]$

9.1.3 $B(270°; -1)$

9.1.4 $0 \leq x < 36,87°$

OR $270° < x \leq 360°$

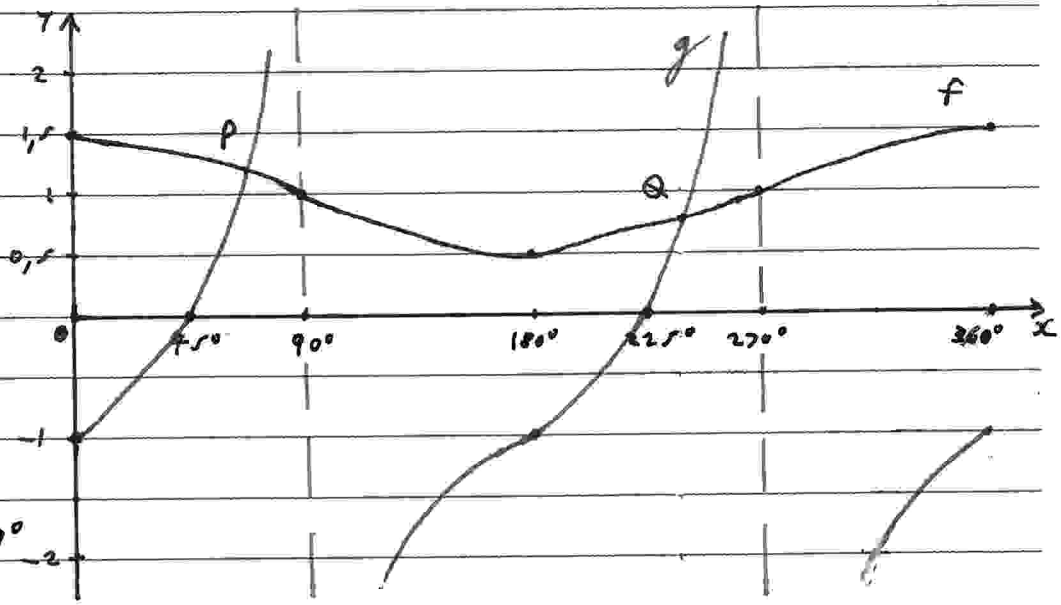
9.1.5 $K < -1$ OR $K > 1$

9.1.6 $90° < x < 270°$

9.2.1 g: x-INT: $y=0$

$\therefore \tan x = 1$

$\therefore x = 45° \text{ OR } 225°$



9.2.2.1 $180°$

9.2.2.2 $y \in [0,5 ; 1,5]$

9.2.2.3 $\frac{1}{2}$

9.2.3 $0° \leq x \leq 65,62°$

OR $90° < x \leq 240,29°$

OR $270° < x \leq 360°$

9.2.4 $-1,5 \leq m < -0,5$ OR $m \in [-1\frac{1}{2} ; -\frac{1}{2})$

9.3.1 $a = 1$; $b = -\frac{1}{2}$ 9.3.2 $P(270° ; -1\frac{1}{2})$

9.3.3.1 1 9.3.3.2 $-1\frac{1}{2} \leq x \leq \frac{1}{2}$

9.3.4 $\sin\theta = \frac{1}{2} \rightarrow \theta = 30°$ OR $\theta = 150°$

9.3.5 $\sin x - \frac{1}{2} = 0 \rightarrow f(x) = 0 \rightarrow T(30° ; 0)$ AND $Q(150° ; 0)$

9.3.6.1 $k > 1\frac{1}{2}$ 9.3.6.2 $k < -\frac{1}{2}$

9.4.1 $m = 2$ AND $n = -1$ 9.4.2 $P(90° ; 2)$ AND $R(270° ; -2)$

9.4.3 $TR = -1 - (-2) = 1$ UNIT

9.4.4 $2\cos x - 1 = 0 \rightarrow \cos x = \frac{1}{2} \rightarrow (60° ; 0)$ OR $(300° ; 0)$

9.4.5.1 $x = 0°$ OR $180° \leq x \leq 360°$

9.4.5.2 $0° \leq x < 24,30°$ OR $245,70° < x \leq 360°$

9.4.5.3 $0° < x < 60°$ OR $180° < x < 300°$

$f(x)$	$g(x)$
+	+
−	−

9.5.1 $m = -1$ 9.5.2 $-2 \leq y \leq 0$ 9.5.3 $n = 45°$

9.5.4.1 $90° < x < 152,04°$ OR $270° < x < 297,97°$

9.5.4.2 $x = 0°$ OR $x = 180°$ OR $x = 360°$

9.5.5 $k = 0$ OR $k = 2$

QUESTION 10

10.1 $AC = 14-2x$, $AB = 6x-10$, $BC = 16-4x$, MIDPOINT THEOREM

$\therefore P = 14-2x+6x-10+16-4x \rightarrow$ PERIMETER $= 20$ UNITS

10.2 $\hat{W} = 16^\circ$, ALT. $\angle$'s $\quad \hat{X} = 70^\circ$, CORRES. $\angle$'s

$\hat{Y} = \frac{1}{2}(180-16-70) = 47^\circ$, $\angle$'s IN Δ AND $=$ $\angle$'s OPP. $=$ SIDES

$\hat{Z} = 180-70-47 = 63^\circ$, $\angle$'s IN Δ

10.3 $F\hat{E}G = \frac{1}{2}(180-4\theta) = 90-2\theta$, $\angle$'s IN Δ; $=$ $\angle$'s OPP $=$ SIDES

$\therefore 6\theta+10+90-2\theta = 180^\circ$, $\angle$'s ON STR. LINE

$4\theta = 80 \rightarrow \theta = 20^\circ \quad \therefore F\hat{E}G = F\hat{G}E = 50^\circ$

$\therefore B\hat{G}H = 50^\circ$, VERT OPP $\angle$'s

$\therefore AB \parallel CD$, CORRES $\angle$'s

10.4.1 IN Δ'S QMR AND QPN

$\hat{Q}$ IS COMMON

$Q\hat{M}R = Q\hat{P}N$, CORRES $\angle$'S $\quad \therefore Q\hat{R}M = Q\hat{N}P$, $\angle$'s IN Δ

$\therefore \Delta QMR \,|||\, \Delta QPN$, AAA

10.4.2 IN Δ'S NRA AND RQM

$RA = MP$, OPP. SIDES OF $\parallel$GRAM. BUT $QM = MP$, GIVEN

$\therefore RA = QM$

$NR = RQ$, MIDPOINT THEOREM

$N\hat{R}A = R\hat{Q}M$, CORRES. $\angle$'S

$\therefore \Delta NRA \equiv \Delta RQM$, SAS

10.4.3 $A = \frac{1}{2}(a+b).h \rightarrow 72 = \frac{1}{2}(PA+MR).6$

$\therefore 24 = 2.PA$, $PA = MR$, OPP. SIDES OF $\parallel$GRAM

$\therefore PA = 12$ cm

BUT $PA = AN$, MIDPOINT THEOREM

$\therefore AN = 12$ cm

QUESTION 11

11.1

$\hat{A} = x$, CORRES. $\angle$'S ; $D\hat{E}A = 90°$, CO-INT. $\angle$'S

$B\hat{E}A = \frac{1}{2}(180 - x) = 90 - \frac{x}{2}$, $\angle$'S OPP = SIDES, $\angle$'S IN Δ

$\therefore B\hat{E}D = 90 - \left(90 - \frac{x}{2}\right)$

$\rightarrow B\hat{E}D = \frac{x}{2}$

(OR)

$\hat{A} = x$, CORRES. $\angle$'S ; $B\hat{C}D = 180 - x$, $\angle$'S ON STR. LINE.

$A\hat{B}E = \frac{1}{2}(180 - x) = 90 - \frac{x}{2}$, $\angle$'S OPP = SIDES, $\angle$'S IN Δ

$\therefore C\hat{B}E = 180 - (90 - \frac{x}{2}) = 90 + \frac{x}{2}$, $\angle$'S ON STR. LINE.

$\therefore B\hat{E}D = 360 - \left(90 + 180 - x + 90 + \frac{x}{2}\right)$, $\angle$'S IN QUAD.

$B\hat{E}D = 360 - 360 + \frac{x}{2} \rightarrow B\hat{E}D = \frac{x}{2}$

11.2.1

$\hat{B}_2 = 45°$, BD BISECTS $A\hat{B}C$.

$\therefore B\hat{E}F = 110 - 45 = 65°$, EXT. $\angle$ OF Δ

$\therefore A\hat{E}D = 65°$, VERT. OPP. $\angle$'S

(OR)

$\hat{A}_1 = 110 - 90 = 20°$, EXT. $\angle$ OF Δ ; $\hat{B}_1 = 45°$, BD BISECTS $A\hat{B}C$

$\therefore A\hat{E}D = 45 + 20 = 65°$, EXT. $\angle$ OF Δ

11.2.2

IN Δ'S ABF AND CDG

$AB = CD$, SIDES OF SQUARE ; $BF = DG$, GIVEN

$A\hat{B}F = C\hat{D}G = 90°$, SQUARE

$\therefore \Delta ABF \equiv \Delta CDG$, SAS $\rightarrow \therefore AF = CG$

11.2.3

$AD = BC$, SQUARE; AND $DG = BF$, GIVEN

$\therefore AG = CF$ $AD \parallel BC$, SQUARE

$\therefore AGCF$ IS A $\parallel$GRAM, ONE PAIR OF OPP. SIDES = AND $\parallel$

(OR) AGCF IS A $\parallel$GRAM, BOTH PAIRS OF OPP. SIDES ARE EQUAL

11.3.1

$B\hat{A}D = 180° - 2Y$, CO-INT $\angle$'S

11.3.2

$A\hat{B}C = B\hat{C}F$, $\angle$'S OPP = SIDES

$\hat{E}_1 = x$, ALT. $\angle$'S

$\therefore E\hat{C}F = x$, $\angle$'S OPP = SIDES

$\therefore \hat{F}_1 = x$, ALT. $\angle$'S

$\therefore A\hat{B}C = B\hat{C}F = \frac{1}{2}(180 - x) = 90 - \frac{x}{2}$, $\angle$'s in Δ

11.3.3 $180 - 2y = 90 - \frac{x}{2} \rightarrow \frac{x}{2} = 2y - 90 \rightarrow x = 4y - 180°$

(OR) $D\hat{C}B = x + 90 - \frac{x}{2} = 90 + \frac{x}{2} \rightarrow 2y = 90 + \frac{x}{2} \rightarrow x = 4y - 180°$

11.4.1 IN Δ'S APQ AND BPR

PQ = PR AND AQ = BR , GIVEN

$\hat{Q} = \hat{R}$, $\angle$'s OPP = SIDES

$\therefore \Delta APQ \equiv \Delta BPR$, SAS

11.4.2 $P\hat{A}Q = P\hat{B}R$, $\equiv \Delta$'s

$\therefore P\hat{A}B = P\hat{B}A$, $\angle$'s ON STR. LINE (EQUAL IN EACH CASE)

11.5.1 IN Δ'S ABC AND EDC

$\hat{A} = \hat{E}$, GIVEN AND $\hat{C}_1 = \hat{C}_2$, VERT OPP $\angle$'s

$\therefore \hat{B} = \hat{D}$, $\angle$'s IN Δ

$\therefore \Delta ABC \,|||\, \Delta EDC$, AAA

11.5.2 $\frac{AB}{ED} = \frac{BC}{DC} = \frac{AC}{EC}$, $|||$ Δ's (SIDES IN PROPORTION)

$\frac{12}{20} = \frac{m}{16} = \frac{6}{n} \rightarrow m = 16 . \frac{3}{5} = 9,6$ cm

$n = \frac{5}{3} . 6 = 10$ cm

QUESTION 12

12.1 $(x+2)\left(\frac{x}{2} + 2\right) = 40 \rightarrow \frac{x^2}{2} + 3x + 4 - 40 = 0$

$x^2 + 6x - 72 = 0 \rightarrow (x+12)(x-6) = 0 \rightarrow x = -12$ (N/A) OR $x = 6$

$\therefore$ DIMENSIONS ARE 6 X 3 METRES

12.2.1 FOUNTAIN: AREA $= \frac{1}{2} . \pi . \left(\frac{6}{\sqrt{\pi}}\right)^2 = 18\ m^2$

GRASS : AREA $= \frac{1}{2} . 6 . 6 = 18\ m^2$

FLOWER BED: AREA $= \frac{1}{2}(3+5).4 = 16\ m^2$

$\therefore$ AREA TO BE PAVED $= 20 \times 15 - (18+18+16) = 248\ m^2$

12.2.2 $90 \times 248 = R\ 22\ 320$

12.3.1 8 BALLS $\rightarrow 8 \times 40 = 320\ mm \rightarrow SA = 2.\pi.20^2 + 2.\pi.20.320$

$SA = 42725,66\ mm^2 \rightarrow SA = 427\ cm^2$

12.3.2 CONTAINER: $V_c = \pi r^2 h = \pi.20^2.320 = 128000\pi$

BALLS: $V_b = 8.\frac{4}{3}\pi.20^3 = \frac{256000\pi}{3}$

$\therefore V(AIR) = 134041,2866\ mm^3 \rightarrow V(AIR) = 134\ cm^3$

12.4.1 $1 = x^3 - y^2x \rightarrow xy^2 = x^3 - 1 \rightarrow y = \sqrt{\frac{x^3-1}{x}}$

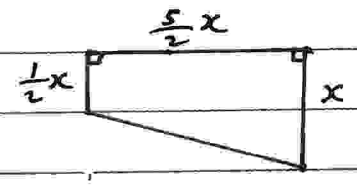
12.4.2 $y = \sqrt{\frac{(1,1)^3-1}{1,1}} = 0,5486\ m \rightarrow y = 54,86\ cm$

13.1.1 SLANT LENGTH $= \sqrt{x^2+25x^2} = \sqrt{26x^2} = x\sqrt{26}$

$\therefore SA = x.2x + 2.\frac{1}{2}x.5x + 2x.5x + 2x.\sqrt{26}x$

$= 2x^2 + 5x^2 + 10x^2 + 2\sqrt{26}x^2 \rightarrow SA = (17+2\sqrt{26})x^2$

13.1.2 THE SLANTED SIDE IS BISECTED BY THE DOTTED LINE, (MIDPOINT TH.) AND THE DOTTED LINE $= \frac{1}{2}x$.



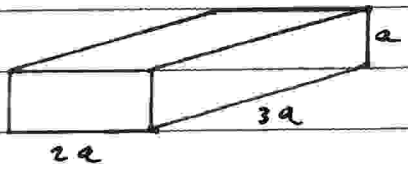
$\therefore V_p = \frac{1}{2}.\frac{5}{2}x\left(x+\frac{1}{2}x\right).2x$

$= \frac{5}{2}x^2\left(\frac{3}{2}x\right)$

$\therefore V_p = \frac{15}{4}x^3$ UNITS^3

13.2.1 $550 = 2(2a^2 + 3a^2 + 6a^2)$

$550 = 22a^2 \rightarrow a = 5\ cm$



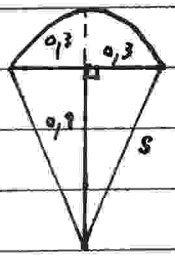
13.2.2 $137,5 = 22.25.k^2 \rightarrow k = \frac{1}{2}$

13.3.1 $V = 20.15.(2,8) = 840\ m^3$ $\therefore$ VOLUME = 840 kℓ

13.3.2 $V(AIR) = 20.15.(0,2) = 60\ m^3 \rightarrow 60000/150 = 400$ PEOPLE

13.4.1 $V_{HS} = \frac{1}{2}.\frac{4}{3}\pi(0,3)^3 = \frac{9}{500}\pi$ AND $V_C = \frac{1}{3}\pi(0,3)^2.(0,9) = \frac{27}{1000}\pi$

$\therefore V_{TOTAL} = \left(\frac{9}{500} + \frac{27}{1000}\right)\pi = \frac{9}{200}\pi\ m^3$



13.4.2 $SA_{HS} = \frac{1}{2}.4\pi(0,3)^2 = \frac{9}{50}\pi$

$s^2 = (0,3)^2 + (0,9)^2$

$\therefore s = \frac{3\sqrt{10}}{10}$

$SA_C = \pi.(0,3).\frac{3\sqrt{10}}{10} = \frac{9\sqrt{10}}{100}\pi$

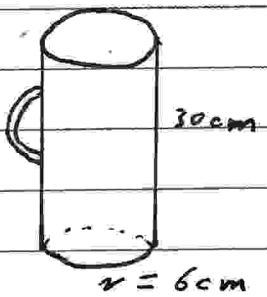
$\therefore SA(TOTAL) \approx 1,46\ m^2$

13.5.1 $V_{JUG} = \pi.6^2.30 = 1080\pi\ cm^3$

$V_{ICE} = 3.4.2 = 24\ cm^3$

$\therefore 1080\pi / 24 = 141,37$

$\therefore$ 141 ICE BLOCKS MAY BE MADE



13.5.2 $(V_{JUG}(\frac{1}{2}\ FULL) = 540\pi)$ NOT REQUIRED

$V_C = \pi.3^2.5 = 45\pi$

$\therefore 45\pi = \pi.6^2.h \rightarrow h = 1,25$

$\therefore$ THE WATER RISES BY 1,25 cm